

1851 Fizeau: Hypotheses on the Luminiferous Ether — verification companion

Hippolyte Fizeau

Fizeau 1851 — verification companion

[!abstract] Legend - □ tip/note — what the source establishes and how it is used - □ quote — verbatim source text

[!success] PDF present `project_sources/1851_Fizeau_Hypotheses_Luminiferous_Ether.pdf` is in this folder.

Bibliographic Info

- **Author:** Hippolyte Fizeau
- **Title:** Sur les hypothèses relatives à l'éther lumineux (on the hypotheses relating to the luminiferous ether)
- **Year:** 1851
- **Publication:** Comptes Rendus de l'Académie des Sciences, vol. 33, pp. 349–355

Claim used in the sheet / notebook

[!note] Background only, not a fitted value Fizeau's moving-water experiment measures the Fresnel drag coefficient $1 - 1/n^2$: light in flowing water is dragged, but only partially. This underpins the water block's $\beta(n - 1/n) \approx 12''$ drag term (Franco slide 13/14) and the statement "light in water = c/n " used as the measured in-medium speed. No number is taken from Fizeau into a sheet cell; it is the experimental warrant for the drag coefficient the derivation uses.

[!warning] Verify against the PDF Confirm the drag coefficient form and that the measured speed is c/n (not the full ether-drag $c/n - v$). The Player-Rogers refinement (group vs phase, n_g vs n_ϕ) that the null-hypothesis argument leans on is a later result, not in Fizeau 1851 — keep those citations separate.

Backlinks

- [[2025_Ligabue_Airy_Relativity]]
- [[1871_Airy_Supposed_Alteration_Aberration]]

Tags

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